

CONSIDERING

- steady-state conditions
- adiabatic
- negligible variation in kinetic and potential energy

THE ENERGY BALANCE PROVIDES

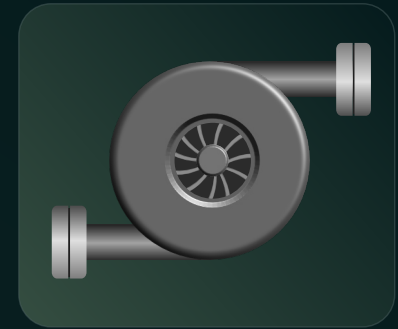
$$\dot{W}_{pump} = \dot{m} (h_2 - h_1)$$

WITH THE ISENTROPIC EFFICIENCY

$$\eta_b = \frac{h_{2,is} - h_1}{h_2 - h_1}$$

WHERE

- η_b , h_1 and $\dot{m}$ are known
- $h_{2,is}$ is calculated with an EOS under isentropic conditions



SYSTEM COMPONENT

P U M P



CONSIDERING

- steady-state conditions
- no work
- negligible variation in kinetic and potential energy

SYSTEM COMPONENT

HEATER/COOLER *

THE ENERGY BALANCE PROVIDES

$$\dot{Q}_{heater/cooler} = \dot{m}(h_2 - h_1)$$

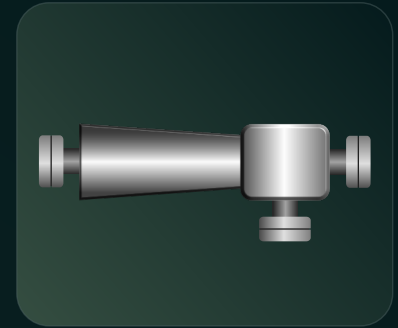
WHERE

- h_1 and $\dot{m}$ are known
- h_2 is calculated with an EOS

*USED IN THE EVAPORATOR,
CONDENSER, AND GENERATOR

MAIN CONSIDERATIONS

- steady-state conditions
- adiabatic
- flow with negligible velocities at the inlets and outlet.
- CPM ejector



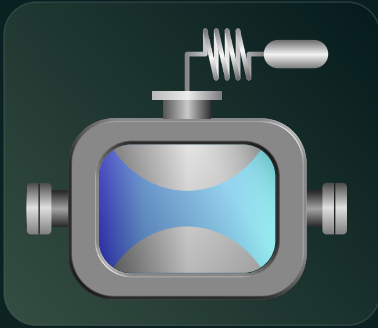
SYSTEM COMPONENT

EJECTOR

MODEL DEVELOPED BY CARDEMIL AND COLLE (2012)

A GENERAL MODEL FOR EVALUATION OF VAPOR EJECTORS PERFORMANCE FOR APPLICATION IN REFRIGERATION. ENERGY CONSERVATION AND MANAGEMENT, V. 64, P. 79-86, SET. 2012.

DOI: [HTTPS://DOI.ORG/10.1016/J.ENCONMAN.2012.05.009](https://doi.org/10.1016/j.enconman.2012.05.009).



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SYSTEM COMPONENT

EXPANSION VALVE

THE ENERGY BALANCE PROVIDES

$$h_1 = h_2$$

WHERE

- h_1 are known
- h_2 (which is equal to h_1) and the pressure at the outlet (which is known) are used by EOS to determine the state of the outlet.